Given a temperature of 201 K, convert to degrees Fahrenheit.

$T_K = T_C + 273.15$

$T_F = \frac{9}{5} T_C + 32$

$T_C = T_K - 273.15$

$T_F = \frac{9}{5}[T_K - 273.15] + 32$

$= -97.87\ ^\circ F$

A metal has a coefficient of linear expansion of $2.6 \times 10^{-5}\ (C^\circ)^{-1}$. Suppose a pipe made of this metal is 1.7 m long. Calculate the temperature increase in C° if its length increases by 7.4 mm.

1D

$\Delta L = \alpha L_0 \Delta T$

$\Delta T = \frac{\Delta L}{L_0 \alpha}$

$= 167.4\ ^\circ C$

$\Delta L = 7.4\ mm = (7.4) \times 10^{-3}\ m$

A metal has a coefficient of linear expansion of 1.1×10^{-5} (C°)$^{-1}$. Suppose a pipe made of this metal is 5.2 m long. Calculate its length increase in mm if it is warmed by 46 C°.

$\Delta T = 46\,°C$

ΔL = _____mm

$\Delta L = \alpha L_0 \Delta T$

$= 2.63\ mm$

Suppose a glass bottle with volume 0.012 m^3 if filled to the brim with a fluid of unknown coefficient of volume expansion. When warmed by 7.1 C°, 16 mL spill onto the lab table. If the glass has linear coefficient 1.9×10^{-5} K^{-1}, calculate the unknown volume coefficient of the fluid in (kC°)$^{-1}$.

(Be careful with units and note that 1000 C° = 1 kC°.)

$16\ ml = 16 \times 10^{-6}\ m^3$

β = _____ (kC°)$^{-1}$

$V_B' = V_B + 3\alpha V_B \Delta T = V_0 + 3\alpha V_0 \Delta T$

$V_L' = V_L + \beta V_L \Delta T = V_0 + \beta V_0 \Delta T$

$V_L = V_B = V_0$

$\Delta V = V_L' - V_B' = \beta V_0 \Delta T - 3\alpha V_0 \Delta T$

$\Delta V + 3\alpha V_0 \Delta T = \beta V_0 \Delta T$

$\beta = \dfrac{\Delta V + 3\alpha V_0 \Delta T}{V_0 \Delta T}$

$= 0.245$

$1\ L = 10^{-3}\ m^3$

$1\ ml = 10^{-6}\ m^3$

A bottle filled with 1.3-L of water falls from a height of 150.2 m and hits the ground ($g = 9.8$ m/s^2). If all the mechanical energy loss of the water went into heating the water, find its **change** in temperature (in C°).

$E = PE = mgh = Q = mc\Delta T$

$\Delta T = \frac{gh}{c}$ $\quad c = 4190 \frac{J}{kg \cdot °C}$

$= 0.351\ °C.$

Suppose 3,141 J of energy are transferred to a block of mass 4.3 kg and specific heat 645 J/(kg·C°). Calculate the temperature change in Celsius degrees.

$Q = m \cdot c \cdot \Delta T$

$\Delta T = \frac{Q}{mc} = 1.13\ °C.$

Suppose transferring 24,512 J to a substance of mass 2.8 kg raises the temperature by 11 C°. Calculate the specific heat of the substance in J/(kg·C°).

$Q = mc\Delta T$

$c = \frac{Q}{m\Delta T} = 796 \frac{J}{kg \cdot °C}$

Suppose 5.6 kg of chloroform at 13.5°C is poured into 9.5 kg of propylene glycol at 36.8°C. Calculate the final equilibrium temperature, neglecting any energy processes associated with mixing, given c_{prop} = 2500 J/(kg·K) and $c_{chloroform}$ = 1050 J/(kg·K).

T = _____ °C

$\Delta T_1 = T_f - T_{ch}^{\circ}$

$\Delta T_2 = T_f - T_g^{\circ}$

$Q_{ch} + Q_g = 0$

$m_{ch} c_{ch} \Delta T_1 + m_g c_g \Delta T_2 = 0.$

$m_{ch} c_{ch} (T_f - T_{ch}^{\circ}) + m_g c_g (T_f - T_g^{\circ}) = 0.$

$$\boxed{T_f = \frac{m_g c_g T_g^{\circ} + m_{ch} c_{ch} T_{ch}^{\circ}}{m_{ch} c_{ch} + m_g c_g}} = 32^{\circ}.2^{\circ}C.$$

An ice cube tray holds 0.29 kg of water at 17°C. How much energy in kJ much be removed from the water to freeze it?

[Use: L_f = 335,000 J/kg; L_v = 2,260,000 J/kg; c = 4190 J/(kg·K).]

$\cancel{Q = mc\Delta T}$

$T_i = 17^{\circ}C$

$T_f = 0^{\circ}C.$

$Q = m c_w (T_f - T_i) - m L_f$

$= (0.29)(0 - 17) - (0.29)(335{,}000)$

$= -117.8$ J.

How much energy (in kJ) is required to convert 2.69 kg of ice at -20°C to water at 16°C?

[Use: c_{ice} = 2100 J/(kg·K); c_{water} = 4190 J/(kg·K); $L_{f,water}$ = 334x10^3 J/kg]

$Q = \underbrace{m_{ice} c_{ice} (T_f - T_i)}_{\text{warm}} + \underbrace{m_{ice} L_f}_{\text{melt phase change}} + \underbrace{m_{ice} c_{water} (T_f - 0)}_{\text{warm}}$

$T_i = -20°$

$T_f = 16°$

$Q = 1,192$ kJ

Suppose 7.2 kg of ice at 0°C is placed in 11.5 kg of water at 28.4°C. Calculate the final equilibrium temperature in Celsius. L_f = 3.35 x 10^5 J/kg; c = 4190 J/(kg·K).

[Hint: There are two possible outcomes - the ice will be either fully or partially melted. Start by figuring out which occurs in this case.]

$Q = m_w C_w \Delta T$ $\quad 28.4 \to 0°$

$= (11.5)(4190)(0 - 28.4)$

$= 1,368,454$ J $= 1.368454 \times 10^6$ J

$Q = m_{ice} L_f = (7.2)(3.35) \times 10^5$

$= 24.12 \times 10^5$ J.

$T_f = 0°$

A window has dimensions 1.6 m × 2 m × 13.9 mm where the latter quantity is the thickness in millimeters. The temperature outside is -17°C, and inside it is 18°C. If the thermal conductivity of the glass is 0.8 W/(m·K), calculate the magnitude of the energy transfer rate through the window in kW.

$R_1 \; R_2 \; R_2 \equiv R = \sum R_i$

H = _____kW

$H = \frac{k}{L} A (T_H - T_C)$

$R = \frac{L}{k}$

$= \frac{A}{R}(T_H - T_C)$

$H = \frac{0.8}{(13.9)\times 10^{-3}} \cdot (1.6 \times 2)(18 - (-17))$

$= 6.45$ kW.

A ball hanging by a wire has a radius of 3.1 m and emissivity 0.63. If the surface temperature of the ball is 92°C and the room is at temperature 28°C, calculate the net rate in kW at which energy is radiated by the ball.

$4\pi r^2$

H = _____kW

$H = A\sigma e (T_H^4 - T_C^4)$

$T_H = (92 + 273.15)K$

$T_C = (28 + 273.15)K$

$H = 41.2$ kW